

NORTHWESTERN UNIVERSITY

Optimizing pulse for gate and decoherence for superconducting qubits

A DISSERTATION

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ABSTRACT

How is it that the sunlight gives us such joy? Why does this radiance when it falls on the earth fill us with the joy of living? The whole sky is blue, the fields are green, the houses all white, and our enchanted eyes drink in those bright colors which bring delight to our souls. And then there springs up in our hearts a desire to dance, to run, to sing, a happy lightness of thought, a sort of enlarged tenderness; we feel a longing to embrace the sun.

ACKNOWLEDGEMENTS

A young scamp about fifteen years old, Isidore Duval by name, and called, for convenience, Zidore, took care of this pensioner, gave him his measure of oats and fodder in winter, and in summer was supposed to change his pasturing place four times a day, so that he might have plenty of fresh grass.

The animal, almost crippled, lifted with difficulty his legs, large at the knees and swollen above the hoofs. His coat, which was no longer curried, looked like white hair, and his long eyelashes gave to his eyes a sad expression.

When Zidore took the animal to pasture, he had to pull on the rope with all his might, because it walked so slowly.

LIST OF PUBLICATIONS

Placeholder: Add your peer-reviewed publications here (published, accepted, and submitted), formatted per your program requirements.

DEDICATION

Placeholder dedication text.

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CHAPTER 1

INTRODUCTION AND BACKGROUND

This is Chapter 1.

We define the cavity resonance angular frequency ω_c (in rad s^{-1}).

This is an attempt at some sort of a thesis template for Northwestern PhD, based on the 2021 guidelines. The guidelines change from time to time, so check. At this time, the page number is required to be within the page margins (harmless warning, aesthetically a little weird).

Cite scipy [1]

Table 1.1: Smoke-test table to verify List of Tables automation.

Quantity	Value
Example	1

Table 1.2: This is a table with a caption.

Something	1	1	1	0
Else	0	0	0	1
What	3			
is this	7			

Figure 1.1 is in a chapter, so it is a figure. However, Figure A.1 is a supplementary figure in the appendix.

This is a temporary holder for a figure. It helps me keep track of figures.

Figure 1.1: **Temporary figure in Chapter 1. a**, A placeholder

CHAPTER 2

METHODS

This is Chapter 2.

We use the total energy decay rate κ (in s^{-1}) to parameterize dissipation.

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Figure 2.1: **Temporary figure in Chapter 2. a**, A placeholder

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Figure 2.2: Smoke-test figure to verify List of Figures automation.

Maupassant, like Flaubert, was a Norman, through his mother, and through his place of birth he belonged to that strange and adventurous race, whose heroic and long voyages on tramp trading ships he liked to recall. And just as the author of “Education sentimentale” seems to have inherited in the paternal line the shrewd realism of Champagne, so de Maupassant appears to have inherited from his Lorraine ancestors their indestructible discipline and cold lucidity.

We denote the dimensionless dispersive coupling parameter as χ (in Hz).

His childhood was passed at Etretat, his beautiful childhood; it was there that his instincts were awakened in the unfoldment of his prehistoric soul. Years went by in an ecstasy of physical happiness. The delight of running at full speed through fields of gorse, the charm of voyages of discovery in hollows and ravines, games beneath the dark hedges, a passion for going to sea with the fishermen and, on nights when there was no moon, for dreaming on their boats of imaginary voyages.

Mme. de Maupassant, who had guided her son's early reading, and had gazed with him at the sublime spectacle of nature, put, off as long as possible the hour of separation. One day, however, she had to take the child to the little seminary at Yvetot. Later, he became a student at the college at Rouen, and became a literary correspondent of Louis Bouilhet. It was at the latter's house on those Sundays in winter when the Norman rain drowned the sound of the bells and dashed against the window panes that the school boy learned to write poetry.

Vacation took the rhetorician back to the north of Normandy. Now it was shooting at Saint Julien l'Hospitalier, across fields, bogs, and through the woods. From that time on he sealed his pact with the earth, and those "deep and delicate roots" which attached him to his native soil began to grow. It was of Normandy, broad, fresh and virile, that he would presently demand his inspiration, fervent and eager as a boy's love; it was in her that he would take refuge when, weary of life, he would implore a truce, or when he simply wished to work and revive his energies in old-time joys. It was at this time that was born in him that voluptuous love of the sea, which in later days could alone withdraw him from the world, calm him, console him.

In 1870 he lived in the country, then he came to Paris to live; for, the family fortunes having dwindled, he had to look for a position. For several years he was a clerk in the Ministry of Marine, where he turned over musty papers, in the uninteresting company of the clerks of the admiralty.

Then he went into the department of Public Instruction, where bureaucratic servility is less in-

tolerable. The daily duties are certainly scarcely more onerous and he had as chiefs, or colleagues, Xavier Charmes and Leon Dierx, Henry Roujon and Rene Billotte, but his office looked out on a beautiful melancholy garden with immense plane trees around which black circles of crows gathered in winter.

Maupassant made two divisions of his spare hours, one for boating, and the other for literature. Every evening in spring, every free day, he ran down to the river whose mysterious current veiled in fog or sparkling in the sun called to him and bewitched him. In the islands in the Seine between Chatou and Port-Marly, on the banks of Sartrouville and Triel he was long noted among the population of boatmen, who have now vanished, for his unwearied biceps, his cynical gaiety of good-fellowship, his unfailing practical jokes, his broad witticisms. Sometimes he would row with frantic speed, free and joyous, through the glowing sunlight on the stream; sometimes, he would wander along the coast, questioning the sailors, chatting with the ravageurs, or junk gatherers, or stretched at full length amid the irises and tansy he would lie for hours watching the frail insects that play on the surface of the stream, water spiders, or white butterflies, dragon flies, chasing each other amid the willow leaves, or frogs asleep on the lily-pads.

The rest of his life was taken up by his work. Without ever becoming despondent, silent and persistent, he accumulated manuscripts, poetry, criticisms, plays, romances and novels. Every week he docilely submitted his work to the great Flaubert, the childhood friend of his mother and his uncle Alfred Le Poittevin. The master had consented to assist the young man, to reveal to him the secrets that make chefs-d'oeuvre immortal. It was he who compelled him to make copious research and to use direct observation and who inculcated in him a horror of vulgarity and a contempt for facility.

Maupassant himself tells us of those severe initiations in the Rue Murillo, or in the tent at Croisset; he has recalled the implacable didactics of his old master, his tender brutality, the paternal

advice of his generous and candid heart. For seven years Flaubert slashed, pulverized, the awkward attempts of his pupil whose success remained uncertain.

Suddenly, in a flight of spontaneous perfection, he wrote *Boule de Suif* (Ball of Lard). His master's joy was great and overwhelming. He died two months later.

CHAPTER 3

CONCLUSION AND FUTURE WORK

3.1 Summary of key findings and significance

3.2 Limitations

3.3 Opportunities for future research

REFERENCES

- ¹Virtanen et al., “SciPy 1.0: fundamental algorithms for scientific computing in Python”, *Nature Methods* **17**, 261–272 (2020).

**OPTIMIZING PULSE FOR GATE AND DECOHERENCE FOR SUPERCONDUCTING
QUBITS**

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Date Approved: August 1, 2021

APPENDIX A
APPENDIX TITLE

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Figure S A.1: **Temporary figure. a**, A placeholder

VITA

Guy de Maupassant was born in 1850 in the Normandy region of France. He attended the University of Caen in Normandy. He volunteered for the Franco-Prussian war and then worked as a government clerk in Paris. Since middle school, he was mentored by Gustave Flaubert. At age 30, he published the “Ball of Lard”, his first major work. In the next decade, he was an extremely productive writer.